3D SMECO PLATFORM

User Manual & System Documentation

Version: 1.0
Project: 3D Smeco platform
Prepared For: End Users, Engineers, Maintenance Personnel, Technical Staff
Technology Stack: Three.js, JavaScript, Vite, GSAP, HTML5, CSS3

3D Smeco platform

1. Introduction

Purpose

The 3D Smeco platform is an interactive 3D visualization system developed for ship engine inspection, analysis, maintenance support, and technical documentation access.

The platform enables users to explore engine assemblies in a realistic three-dimensional environment while providing direct access to technical documentation and component information.

The goal of the system is to improve maintenance efficiency, reduce troubleshooting time, and provide a modern digital representation of complex marine machinery.

2. System Overview

The application consists of four primary modules:

1. Interactive 3D Viewer

Provides a real-time visualization of the engine and its assemblies.

2. Hierarchical Navigation System

Allows users to navigate through engine systems and subassemblies.

3. Information & Documentation Panel

Displays component information and related technical documentation.

4. Maintenance Support Tools

Provides access to maintenance procedures, schematics, and exploded views.

3. Main Features

The platform provides the following capabilities:

- Interactive 3D engine visualization
- Real-time camera navigation
- Component selection
- Automatic focus mode
- Hierarchical system navigation
- Technical documentation access
- Exploded view inspection
- Visibility control
- Modern responsive interface
- Cross-platform compatibility

4. User Interface

4.1 Main Viewer

The central area of the application contains the interactive 3D model.

Users can:

- Rotate the model
- Zoom in and out
- Pan the camera
- Select individual components
- Inspect assemblies

4.2 Hierarchy Panel

The hierarchy panel is located on the left side of the screen.

Its purpose is to provide structured navigation through the engine systems.

Example:

GLOBAL

1 Structure
2 Exhaust System
3 Plate Heat Exchanger
4 Port Generator
5 Main Engine No1
6 Transmission
7 Pipes
8 Valves
9 Propeller
10 Fire System
11 Lube Oil Tank
12 Service Air Receiver
13 Duplex Oil Strainer

Each category may contain multiple subassemblies and individual components.

4.3 Information Panel

When a component is selected, the information panel displays:

- Component name
- Description
- Technical documentation
- Schematics
- Maintenance procedures

The information panel appears automatically after a component is selected.

5. Navigation Controls

Mouse Controls

Rotate Camera

Hold the left mouse button and move the mouse.

Pan Camera

Hold the right mouse button and move the mouse.

Zoom

Use the mouse wheel.

Select Component

Left-click on a component.

6. Hierarchy Management

The hierarchy system organizes engine components into logical groups.

Example:

5 Main Engine No1

51 Main Engine No1
52 Gearbox
53 Filters
54 Fuel Oil Filter No1
55 Main Engine No1 Base

Users may:

- Expand categories
- Collapse categories
- Hide categories
- Isolate categories
- Focus on specific assemblies

7. Component Focus Mode

Focus Mode allows the user to isolate a selected component.

Activation

Focus Mode is activated by:

- Clicking a component in the hierarchy
- Clicking a component directly in the 3D scene

Behavior

When Focus Mode is enabled:

- The camera automatically moves to the component
- Other components are hidden
- The information panel is displayed
- Documentation becomes available

Benefits

- Improved inspection
- Easier maintenance planning
- Faster troubleshooting
- Better component understanding

8. Documentation System

The platform supports direct access to technical documents.

Supported document types:

- PDF Specifications
- Maintenance Manuals
- Schematics
- Technical Drawings
- Inspection Procedures

Accessing Documentation

1. Select a component.
2. Open the Information Panel.
3. Click the Documentation button.

The document will open in a new browser tab.

Example:

C32_1000_HP_SPECIFICATIONS.pdf

9. Exploded View

The Exploded View feature separates assemblies into individual parts.

Purpose

Exploded View helps users:

- Understand assembly structure
- Identify components
- Analyze mechanical relationships
- Prepare maintenance procedures

Usage

1. Click the "Explode" button.
2. The model expands automatically.
3. Inspect components.
4. Click again to return to the original assembly.

10. Visibility Management

Each category contains visibility controls.

Users can:

- Show components
- Hide components
- Display specific systems only

This feature is useful when working with large and complex assemblies.

11. Maintenance Workflow

Recommended workflow:

Step 1

Locate the required system through the hierarchy.

Step 2

Select the component.

Step 3

Activate Focus Mode.

Step 4

Open technical documentation.

Step 5

Review maintenance procedures.

Step 6

Perform inspection or maintenance.

12. Troubleshooting

Model Does Not Load

Possible causes:

- Missing model files
- Network issues
- Browser caching problems

Recommended actions:

- Refresh the page
- Clear browser cache
- Verify model availability

Documentation Does Not Open

Possible causes:

- Missing PDF file
- Incorrect document path
- Browser popup restrictions

Recommended actions:

- Verify document location
- Check browser settings
- Confirm PDF availability

Performance Issues

Possible causes:

- Large model size
- Insufficient hardware resources

Recommended actions:

- Close unused applications
- Use hardware acceleration
- Update graphics drivers

13. System Requirements

Recommended Hardware

Processor:
Modern multi-core CPU

Memory:
8 GB RAM minimum
16 GB RAM recommended

Graphics:
Dedicated GPU recommended

Storage:
500 MB free space

Supported Browsers

Fully supported:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

14. Technical Information

Frontend Technologies

- Three.js
- JavaScript (ES6)
- HTML5
- CSS3
- GSAP Animation Library

Build Environment

- Vite

Supported Assets

- GLB Models
- HDR Environments
- PDF Documentation

- Images
- JSON Data

15. Best Practices

For the best experience:

- Use the hierarchy panel for navigation.
- Use Focus Mode when inspecting components.
- Keep documentation updated.
- Verify component naming consistency.
- Regularly update browser versions.
- Use Exploded View during maintenance planning.

Conclusion

The 3D Smeco platform provides a modern and efficient environment for engine visualization, inspection, and maintenance support.

By combining interactive 3D technology, structured component hierarchy, and integrated technical documentation, the platform enables engineers and maintenance personnel to access critical information faster and perform technical tasks more effectively.

The system is designed to improve operational understanding, reduce maintenance time, and support the digital transformation of marine engineering workflows.